

Section 1

NEW PRODUCTS IN A CHANGING WORLD

1

ROBOTS ARE CHANGING THE FACE OF CUSTOMER SERVICE

by Alicia A. Grandey and Kayley Morris

In the *Star Wars* franchise, C-3PO is a protocol droid that serves on the front lines of galactic war, demonstrating advanced knowledge of etiquette across cultures and an ability to speak more than 7 million languages.

Though this depiction of a robot assistant is fiction, it's not far from the robots that we see assisting on a different type of front line: the front lines of customer service. Robots like Hilton's "Connie" and Softbank's "Pepper," though not quite as advanced, utilize their abilities in language and navigation to create better guest experiences in hotels, restaurants, and shops.

Maybe you think you haven't interacted with a service robot, but you've most likely used a version of them: the self-service kiosks at the grocery store. Thanks to the Covid-19 pandemic, such technology has been on the rise in the past few years. Service interactions evoked fears of viral contagion and "maskual harassment," so self-service kiosks were thrust front and center into the trenches of customer service. Now as the pandemic fears wane, service kiosks are here to stay and robots are the future of service.

The benefits of service robots are clear. They won't spread airborne viruses or get burned out from harassment. They have the potential to reduce costs, improve efficiency, and automate tedious tasks. Outside of the occasional glitch or software update, robots are available to work 24/7 without sick leave, holidays, or PTO, guaranteeing that the hotel or car-rental front desk is always staffed for customers' convenience.

But to see these gains, *service robots must be designed and implemented the right way*, otherwise customers—and human coworkers—will avoid interacting with them. Emerging evidence highlights the ways that robots can work best in customer service.

Robots Designed to Be Humanlike—to a Point

Customers have expectations for interpersonal rapport and “service with a smile”—so can a service robot satisfy customers?

Service robots that can be given human attributes (“so cute!”)—including emotional qualities—are more likely to satisfy customers. A 2022 study in the *International Journal of Hospitality Management* found that a more humanlike robot was more satisfying to customers *only* if perceived as female, consistent with stereotypical expectations for interpersonally oriented traits.¹

To be clear, “humanlike” doesn’t mean that service robots require expressive faces and human bodies. Consider how *Star Wars*’ robots R2-D2—and more recently BB-8—could elicit laughter and sympathy despite having no faces, limbs, or voices. And a robot that is too humanlike can feel eerie and create discomfort, known as the uncanny valley effect (see the movie *Polar Express*).

More important than designing robots to *look* human is for customers to *perceive* them as emotional beings. Using the first robot-staffed hotel in Japan for context, a 2021 study published in the *Journal of Applied Psychology* asked 194 hotel guests at checkout about their satisfaction with their stay.² At check-in, half of the guests were given instructions to anthropomorphize the hotel robots—to imagine them as beings who could think and feel—and the other half did not get these instructions. Though both groups interacted with the same service robots, those who humanized the robots had higher satisfaction—and this was due to perceiving the service robot’s capacity for *feelings* more than for thinking. Hotel guests were also more forgiving of service failures when they ascribed feelings to the robots than if they did not. After all, to err is human—such that customers may view failures with humanlike robots with more empathy. In other words, if service robots are still in the beta version, make sure customers see them with empathy—just as they would new employees who are still in training.

So replacing a self-serve kiosk with a service robot is not enough for customer satisfaction—customers want an emotional connection. This doesn’t require a perfect performance or eerily human expressions—just a robot name tag that says “Jennifer” and a placard that tells customers to be considerate of their feelings is sufficient.

Robots Designed for Functionality—Not Just Novelty

While still emerging in the United States, robots are more common in Asia. An analysis of online satisfaction ratings from hotel guests showed that interacting with a service robot evokes mostly positive emotional reactions, such as around the anthropomorphized “cute” robot as a welcome agent and the surprise and delight (especially for children) of a robot delivering room service.³

Novelty plays a big role in customer perceptions of service robots, but will they continue to satisfy once every cash register and front desk has one? Beyond the novelty, the hotel guests studied also liked their functionality.

Functionality is by far the most important aspect of human-robot interactions and the greatest determinant of customer experience. Placing robots in service positions often gives people false hope for efficient and error-free performance, but as with any technology, glitches and user error can frustrate the experience.

We can learn from the case of self-checkout kiosks, which were also originally intended to improve efficiency and reduce costs. Big box stores like Walmart and Sam’s Club, as well as Panera Bread, have been replacing cashiers entirely. Unfortunately, as they became ubiquitous, self-checkouts slowly became hated by customers due to frustration with using the technology and irritation at the loss of services (bagging groceries).

Call centers that deal with high volumes and routine requests have long used chatboxes and automated systems to assess the emotional tone of a customer’s voice. As reviewed in 2018 in *Harvard Business Review*, Aida is a virtual assistant at a Swedish bank that can help with simple transactions,

leaving less monotonous tasks for the humans.⁴ Aida can also tell if customers are frustrated and can send the person over to a human if the service issue cannot be resolved effectively.

However, the human service employees should not be invited to interact solely with abusive customers, which would induce burnout and turnover. Ideally, a function of the service robot could be as protection—a “bouncer,” if you will—that blocks abusive or derogatory customers from human agents. The functionality to differentiate the technologically frustrated or fearful customer from the abusive one can help human employees welcome the robot to the team.

Over time, as service robot technologies become more widespread and integrated into everyday life, their novelty will wear off. To ensure the long-term success of robots in customer service, it is important to strike a balance between utilizing their benefits and maintaining a human touch.

Robots Matched to Customer Base and Tasks

Even if you have anthropomorphized and highly functional robots, your customer base must accept robot service interactions, which require both a psychological and technical readiness. While humanoid robots have long been present in Asia, customer acceptance is still a major obstacle in the United States.

Are robots right for your company? Industries that have highly personalized client services that require a rapport, trust, and problem-solving are probably not a great place for service robots. Services that are more standardized and automatic are a better fit, such as cash register transactions and hotel check-ins and errands.

Yet the nature of the customer base also matters. In order to have a successful interaction with a robot, customers must have both the confidence and the desire to interact with the robot. While hotels see a lot of success with concierge robots, high-end clients may demur to invite a robot to come to the hotel room with extra towels and prefer a human voice or face. The demographic that's most confident and ready to accept robots is a predominantly young and male consumer base. Other customers may doubt their own capability to interact or may not believe that the robot is functionally able to assist them. Companies also must consider where tech can become more cumbersome than helpful if the robot is not able to help the majority of customers effectively.

To manage the customer-robot service interactions, human employees are essential. They need to be present to guide customers through the robot interactions as needed. To avoid being like the self-checkout lane at the grocery store, don't simply put your customers to work and remove the option for human services completely. In short, robots and coworkers are the most effective service team.

Robots Introduced as Coworkers—Not Replacements

Introducing robots to customer service front lines can feel relieving *and* threatening to human employees. Putting robots front and center can reduce monotonous tasks and customer mistreatment, but human coworkers may be skeptical of robots' functionality—or may fear they are too functional and will steal their jobs.

Be clear that human employees are an essential part of the successful integration of new technologies into a business. They help anthropomorphize the robots for customers (for example, “Have you met Jennifer, my new robot coworker?”), they ensure robot functionality and take over when not meeting expectations, and their own comfort interacting with the robot serves as a role model for customers who might otherwise be skeptical.

To increase comfort with robot teammates, managers should communicate that the goal is not to replace humans but to successfully integrate robot and human labor for the optimal customer experience. Managers help by effectively communicating about the robot technology—its functionality and its

limits—and offering reward programs for technology training and expertise. In addition, managers should communicate how the new technology can protect employees from the most tedious tasks and unpleasant customers—and might even be fun and enjoyable to interact with.

While robots are likely to take over some frontline jobs that are standardized and routine—the grocery cashier or hotel front-desk clerk, for example—the diversity and complexity of human nature will still require humans to be involved in even these forms of service. When employees realize that their tasks become more interesting with the robot dealing with the monotonous tasks or abusive customers, they might just want to include that robot in the next happy hour invitation.

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Service robots are an emerging technology, and they might have unexpected benefits beyond the customer service interaction. Management professor Pok Man Tang, one of the authors of the 2021 study in the *Journal of Applied Psychology*, has discovered early evidence that interactions with anthropomorphized service robots enhance customers' openness to diversity: both their acceptance of nonconventional products and attitudes toward minoritized service employees.⁵

Ultimately, whether people continue to enjoy robots in customer service will depend on the interpersonal skill and humanness (not too little or too much) and functionality of the technology, and the acceptance of the humans (customers and coworkers). Robot technology should not simply be added as a novelty but carefully integrated to deliver value to customers and support employees—maintaining a balance between automation and human interaction.



Service robots have the potential to reduce costs, improve efficiency, and automate tedious tasks. But to realize these gains, they must be designed and implemented in the right ways. If not, customers—and human coworkers—will avoid interacting with them.

- # Once the novelty of service robots fades, functionality is by far the most important aspect of customer experience.
- # Service robots that are given human attributes—including emotional qualities—are more likely to satisfy customers, but customers will be turned off by robots with eerily humanlike expressions.
- # Industries that have highly personalized client services that require rapport, trust, and problem-solving are not a good match for service robots. More standardized and automatic services are a better fit.
- # To increase employee comfort, managers should communicate that the goal is to integrate robot and human labor for optimal customer experience, not to replace humans.

NOTES

1. Soobin Seo, “When Female (Male) Robot Is Talking to Me: Effect of Service Robots’ Gender and Anthropomorphism on Customer Satisfaction,” *International Journal of Hospitality Management* 102 (April 2022), <https://doi.org/10.1016/j.ijhm.2022.103166>.

2. Kai Chi Yam et al., “Robots at Work: People Prefer—and Forgive—Service Robots with Perceived Feelings,” *Journal of Applied Psychology* 106, no. 10 (2021): 1557–1572, <https://doi.org/10.1037/apl0000834>.
3. Cheng Chang et al., “Factors Influencing Consumers’ Willingness to Accept Service Robots: Based on Online Reviews of Chinese Hotels,” *Frontiers in Psychology* 13 (October 2022), <https://doi.org/10.3389/fpsyg.2022.1016579>.
4. H. James Wilson and Paul R. Daugherty, “Collaborative Intelligence: Humans and AI Are Joining Forces,” *Harvard Business Review*, July–August 2018, <https://hbr.org/2018/07/collaborative-intelligence-humans-and-ai-are-joining-forces>.
5. Yam et al., “Robots at Work.”

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2

BIOMETRICS ARE BECOMING A BUSINESS REALITY

by Therese Stowell

It's a tough time to be a business competing on customer experience. Digital-first disruptors are now firmly embedded across multiple industries, spelling trouble for incumbents as consumers compare these brand relationships to those with tech-centric companies. Today, consumer expectations are changing as we become familiar with quick access to services, such as making a purchase with just a few clicks on Amazon, banking remotely with Revolut, or ordering food and groceries using Deliveroo. These digital-first businesses have built their processes with the customer at the center. Given how nimble they are, that's been a difficult act to follow.

Covid-19 provided traditional businesses with an opportunity to level the digital playing field. Lockdowns and other restrictions were pivotal in accelerating digital transformation as companies were forced to move online or be left behind. The ensuing development of digital services, apps, and online platforms was not only what customers needed, but a commercial necessity. Suddenly, businesses that did not typically have digital at their core had an opportunity to prioritize this and achieve the gold standard of customer experience set by the disruptors and tech giants.

These digital services are often accessed via smart devices for convenience and speed. And now, driven by the need to enhance the consumer experience, they are incorporating new technologies, such as biometric identification (fingerprint or facial recognition), to support easy access and secure customer data.

Earning Customer Trust

Among the digital-first businesses, fintechs such as Revolut and Monzo exploded into the marketplace. They changed the parameters of customer experience and user convenience by offering simpler access to online banking products and services than their competitors.

The challenge they faced was how to verify users' identities remotely—something a bank teller would typically do in-branch—with a process that was compliant with regulatory requirements and user-friendly, but also prevented fraud. This paved the way for ID and biometric verification, allowing customers to prove who they say they are and bank online without needing to go in-branch or in-store. Traditional banks eventually followed suit, prompted by evolving customer expectations and the restrictions of the pandemic.

Soon, other regulated industries that stood to benefit from meeting their users online, such as gambling and on-demand delivery, looked to fintech's example and adopted the same methods to have high assurance of identity and age verification. For consumers, familiarity with digital services had grown so much during lockdowns that most were comfortable with biometrics as an authentication method. In fact, data from Onfido showed that nine out of 10 consumers felt comfortable accessing services online.¹

For example, car-sharing service Drivy uses facial biometrics to verify drivers digitally, meaning that car owners no longer have to meet the renter in person. The positive customer experience resulted in a 38% increase in users verified and onboarded to the Drivy platform. It also reduced car theft because a user's identity is tied to the vehicle for the duration of the rental. Other mainstream rental companies like Hertz and Avis are now following suit, allowing renters to go straight to their cars and skip the lines.

Biometrics are also making waves in the hospitality sector. Platforms like Sidehide enable hotel customers to book and check in via an app, using biometrics rather than handing over documentation upon arrival. That's not to say that tech giants aren't innovating too—early trials of Amazon One in the United States are enabling contactless retail payments in supermarkets like Wholefoods using the customer's palm.

When Friction Makes Sense

When we think of the best customer experiences, they mostly center around how quickly and easily products and services can be accessed. In many cases, this is one of the primary motivations for businesses integrating biometrics. But speed isn't everything. It's important to acknowledge that a smooth customer experience doesn't just mean how quickly customers can achieve what they want to achieve; it's also about privacy and trust.

The growth of online traffic brought new opportunities for fraudsters and hackers of all abilities. In fact, less-sophisticated fraud—in which doctored identity documents are readily spotted—jumped by 37% year on year in 2022, showing that bad actors are focusing on broad, brute force attacks. In other words, quantity not quality.

Whether through online identity fraud or data breaches, the financial services sector has always been the primary target for cybercriminals; the industry has become very familiar with these attacks and is well prepared to deal with them. That doesn't stop fraudsters from trying—attempts faced by financial services businesses increased by 23% last year. In regulated industries like these, biometrics play a vital role in fraud prevention, a vital component of building customer trust.

However, other industries newer to digital services have had to adapt. Trust-based services like dating apps are increasingly using biometrics as a competitive advantage; ensuring users match their profiles with other verified users gives them more confidence in who they are meeting and reduces fraud. In this way, businesses can use the technology to offer trust and safety.

The relationship between customer experience and security is a delicate one. Naturally, everyone wants to know that their personal information and money are secure. But somewhat counterintuitively, when the stakes are high, the right amount of friction in the customer journey can make users feel safe. Biometrics are increasingly the method selected to strike this balance—adding an additional layer of security that is low effort for the consumer and quietly reassuring to see.

Biometrics Aren't for Everyone

That's not to say that biometrics are a silver bullet. Typically, they're a more appropriate solution for highly regulated industries or businesses that need to verify customers deemed as potentially high-risk. They're sometimes used for purposes of age verification too, but facial biometrics on their own can be unreliable unless there is a legitimate ID to compare it against.

Because digital services are so commonplace, and humans can't possibly manually verify every customer interaction with a digital service, there is a greater role for AI to play in automating the biometric verification process. This comes with its own risks. Developing AI to recognize people as real humans means training algorithms on data that is representative of the population. If it isn't, businesses risk a biased product that could exclude people and groups. Companies introducing biometrics into their customer experience must ensure that the AI supporting them is developed ethically and based on representative data.

And while applications of the technology are now widespread, concern over how and where customers' biometric data is stored is also a barrier to adoption for some. Both brands and the tech industry are working to address this. One example is introducing the concept of reusable or shareable identity as a version of biometrics where identity information is stored on a consumer device rather than being held by businesses.

Making Biometrics a Reality

Where they are viable, biometrics can underpin a smoother, more secure customer experience. Businesses adopting biometrics often do so to support a customer base with significant scale, meaning AI is likely to play a role in managing the verification process. While AI has high potential to amplify human biases, how a business integrates it can make or break the biometrics experience. This means building algorithms using diverse data sets to ensure inclusivity and accessibility for the entire population. Monitoring known algorithmic biases around metrics like gender, race, and age can bolster more equitable access to services globally.

Biometrics also need to be easy for people to use. Just like any website or digital service, a smooth user journey is fundamental to ensure biometrics are accessible to all. It's not enough to simply add an innovative technology into the mix and assume that will be enough to improve experience.

The expectations for brand customer experience are higher than ever, particularly as digital channels have become the first point of engagement, feeding into a culture of on-demand access. As the incidence of fraud simultaneously rises, estimated to cost businesses \$5.38 trillion annually, biometrics have emerged as a way to strike the balance between speed and security. They're giving businesses the opportunity to achieve "Amazon-like" levels of digital customer experience—if they can be introduced ethically.



TAKEAWAYS

Businesses often utilize biometric customer verification processes to enhance speed and reduce friction. But a smooth customer experience doesn't only hinge on how quickly customers can achieve what they want; it's also about earning customer trust.

- # Biometrics—such as facial or fingerprint recognition—are increasingly the selected method to strike a balance between speed and privacy. They add an additional layer of security that is low effort for the consumer and quietly reassuring to see.
- # Biometrics must be easy to use. Businesses should not simply add innovative technologies into the mix without considering the impact on customer experience.
- # There is a greater role for AI to play in automating the biometric verification process, but such options come with their own risks, which must be managed.

Editor's note: Revolut, Drivy, and Sidehide use Onfido's services.

NOTE

1. Onfido, "User Research: Digital by Default," <https://onfido.com/report/digital-by-default/>.

Adapted from "How Biometrics Are Transforming the Customer Experience," on hbr.org, March 29, 2023 (product #H07JQP).

3

HOW EARLY-ADOPTER COMPANIES ARE THINKING ABOUT APPLE VISION PRO

by Cathy Hackl

The Apple Vision Pro recently launched in the United States to a mix of both fanfare and skepticism. From skeptics, the arrival of Apple's new device—priced at \$3,500—has been met with questions like “Who is this for?” and “What’s the point of it?”

To understand the hype, it's important to have context: For many, the headset is the biggest test yet of a new field of technology, called spatial computing, that proponents believe could usher in a new era of computing. Business leaders should take note of the Apple Vision Pro not for the device itself, but for the spatial computing capabilities that come with it and the possibilities that may lay ahead for their businesses and products.

Spatial computing uses AI, computer vision, augmented and virtual reality, and other technologies to blend virtual experiences into a person's experience of the physical world and create a form of computing that works in three dimensions. Driven by advances in software, hardware, data, and connectivity, it enables people to interact with each other and tech in new ways, and enables machines with new capabilities to navigate our physical world. By adding a virtual layer to a user's experience of the physical world, it allows images, sounds, and accelerations processed by a device to contribute to real-time perceptions. Our eyes become a mouse, and our gestures are clickers.

From a business perspective, it has the potential to expand computing into everything you can see, touch, and know, promising the kind of change that came with the introduction of the smartphone era.

So what does this mean for businesses? And how are companies trying to engage with this new frontier? Early adopters of Apple's new product—and other spatial computing devices—can give businesses some hints of its potential, and what may or may not work. Businesses that are building native apps for visionOS, the operating system that powers the Apple Vision Pro, are already learning key lessons.

Who Are the Early Adopters and What Have They Learned?

When the Vision Pro was released, companies from Lowe's to e.l.f. Cosmetics launched native apps to go with it. Other early-adopter brands, like Hanifa, are already working on creating new customer experiences. These native visionOS apps give us a glimpse into how spatial computing could change how business, customer service, entertainment, and work will evolve.

Lowe's

Lowe's launched Lowe's Style Studio, an app that uses the immersive, spatial computing abilities of the Vision Pro to allow users to create, explore, and bring their kitchen renovation to life. Instead of starting their renovation by looking at 2D screens or flat images, homeowners can step into an immersive 3D kitchen scene. After finalizing their renovation plans, customers can share their style boards with significant others, contractors, or interior designers.

According to the company, its app is built to leverage Apple Vision Pro's unique capabilities like its true depth camera and gesture tracking.

"Our lives and homes are inherently three-dimensional, yet the tools we've had for designing and visualizing these spaces have largely been limited to 2D representations on smartphones or computers. These traditional methods lack the depth and realism needed to understand a space," said Seemantini Godbole, executive vice president, chief digital and information officer at Lowe's.

Godbole added that launching Lowe's Style Studio on Apple Vision Pro was about keeping pace with technological advancements and meeting customers' evolving expectations by investing in emerging technologies to eliminate the friction in home improvement.

Building an app for the Vision Pro—a technology it hadn't worked with before—in time for launch day required Lowe's to embed members of its mobile technology team into its innovation labs team. That "enabled us to complement our lab team's expertise in 3D visualization, spatial computing, and game design with our mobile tech team's expertise in iOS UX and UI design and deep day-to-day familiarity with Lowe's omnichannel business," said Godbole.

The brand sees a business advantage to being early to spatial computing. It understands home improvement as a unique category within retail—inherently complex, often requiring specialized knowledge, expertise, and a deep understanding of spaces, and potentially deeply personal. Spatial computing, it believes, may help their customers manage the complexity of design and make better home improvement decisions.

e.l.f. Cosmetics

The cosmetics company e.l.f. launched "your best e.l.f." on the Vision Pro, making it one of the first beauty brands to do so. The company has already explored new opportunities for its brand with experiences in Roblox and an active presence on Twitch and TikTok. It saw the opportunity to launch on the Apple Vision Pro as a way to use spatial computing and chart a new path in consumer engagement. "Testing and learning is in our DNA, and in order for us to create experiences on Vision Pro, we have to test and learn and tailor experiences to the community," said e.l.f. Beauty's chief digital officer Ekta Chopra.

When it comes to the business advantage of being early, Chopra indicated that the company sees spatial computing as part of the brand's journey and vision to disrupt norms, shape culture, and connect communities. "There's even more opportunity in spatial computing because if your community is living

in that device for hours, you go with them on the journey. For beauty brands, it creates an experience that goes so much further than just a product, creating an experience for users to connect with your whole self,” she said.

PGA Tour

The PGA Tour released the PGA Tour Vision app for the Vision Pro. The app includes a 3D spatial map of the par-3 seventh hole at Pebble Beach Golf Links, the site of the AT&T Pebble Beach Pro-Am. The app will update weekly to include additional holes leading up to The Players Championship. Scott Gutterman, PGA Tour senior vice president, digital operations, said in a press release, “PGA Tour Vision for Apple Vision Pro brings golf fans inside the ropes and directly onto the tee boxes and greens of the world’s most iconic courses, no matter where they are.”

The app uses 2D video and 3D-rendered models generated from real-time live-shot data captured during the tournament. This is where spatial computing comes in, rendering 3D models using live data in real time. “We were able to take the learnings from past AR/VR/XR experiences and put them into practice with PGA Tour Vision,” said Eric Hanson, vice president of digital product development.

“The main goal is to grow our fan base, and this platform presents an opportunity to reach fans in a new manner,” said Hanson. “A golf tournament is played across 100 to 200 acres, and we change courses every week. It’s hard for a fan to get a sense of what it is like to play one of these courses because you can only see a little bit of each hole on TV. Spatial computing allows fans to go inside the ropes of the competition and see what it is like to play each hole. In the immersive mode, you’re placed on each tee box so you can see what the players experience when they take their first tee shot.”

Hanifa

Hanifa first became a well-known fashion name during the 2020 pandemic after launching the first virtual fashion show. Walking pantsuits and fashionable outfits walked across the screens of those who watched, becoming a viral sensation across the internet and the fashion world. “As a fashion brand whose ethos is tied to pushing fashion’s boundaries, we want to bring fresh new ideas to life in the Apple Vision Pro,” said Anifa Mvuemba, who founded the luxury brand.

While Hanifa’s spatial computing app is still in development, the company aims to be an early adopter. “Our efforts involve the integration of digital elements into physical spaces, offering new possibilities for immersive experiences and interactions,” said Mvuemba. This includes enhancing customer experiences, creating virtual showrooms, and showcasing products.

These are, of course, experiments, and it’s too early to know what will work and what brands and businesses need to know to make spatial computing work best for consumers. But there are overlaps to how these companies are approaching this task. For one, they’re thinking about how a spatial computer acts as an agent for its wearer. While developers must create apps that support depth, gestures, effects, and immersive scene types, they also must take into account who will be experiencing their app and how they wear the Vision Pro.

Barriers to Entry

At launch, Apple announced that more than 600 apps had been built specifically for the Apple Vision Pro and would be available in the App Store. However, adapting regular iOS apps to the Apple Vision Pro from mobile is no simple task. Apple created an all-new platform to develop on the Vision Pro

called visionOS Simulator. Developers have the option to develop in windows, volumes, or spaces. This creates an all-new dimension for developers to consider when creating applications for the Apple Vision Pro.

Developers face challenges from designing for hand gesture input all the way to gaps in team-based knowledge. As far as hardware is concerned, developers will need a MacBook Pro with at least an M2 Apple Silicon chip. Plus, there is the cost of the Vision Pro itself. Companies will need to decide whether they're ready to embrace native 3D and spatial content, and whether investing in an app for the first Apple Spatial Computing device will be a net benefit for R&D.

Barriers to entry to develop spatial computing apps for the Vision Pro include:

- **Technological expertise:** Finding developers with the knowledge, skills, and expertise in spatial computing, development knowledge of game engines like Unity, or visionOS could pose a challenge or expense for new entrants.
- **App competition:** Early adopters of the Apple Vision Pro are the first to gain market share on the new platform. They have the chance to learn early and adapt to changes in the market. However, businesses that wait and learn from the first iteration of the Vision Pro may make up for a later entry with established best practices, seasoned developers, and a larger user base of Vision Pro owners. The Vision Pro may have launched with 600 apps but there are 1.8 million apps on Apple's App Store. This might become more relevant when future iPhones support more spatial capabilities or in future iterations of the Apple Vision Pro, or if more platforms like Meta start supporting spatial video and spatial experiences that could be ported.
- **Discovery and visibility:** Apple has projected it could sell as many as 350,000 units of the Vision Pro in its first year.¹ Compare that to the 200 million iPhones Apple typically sells annually. The up-front costs and investment in development teams for the Vision Pro should be considered when choosing to plan, develop, and market an app for the Vision Pro.
- **Platform lock-in:** Developing an app exclusively for visionOS could mean the app is locked into the Apple ecosystem. Other head-mounted displays like the Meta Quest are on track to roll out spatial video, but app support might emerge more slowly. Business professionals should consider where their target audience is and where they will start to migrate to. Over time, key audiences may be on multiple devices or spatial computing experiences may become cross-platform.

As with any new technology, the biggest barrier to technology is figuring out the best way to use it. The Apple Vision Pro offers new features—ones that are 3D, spatial, and immersive. It's up to the early adopters and their developers and partners to figure out the best practices, implementations, and creativity to make apps that will be a wearer's first pick.

What Is the Business Value of Being Early?

Being an early adopter of spatial computing benefits businesses in several ways. Those companies that dive in and create spatial computing apps for the Vision Pro today give themselves a competitive advantage and position themselves as leaders in their industries.

They have a chance to influence the development of this new technology field, enact best practices, and shape the early development and monetization of customer experiences. Early adopters of the Vision Pro can build brand loyalty in a new space on a new device. This is a chance for companies that may have struggled with mobile or social media to get a head start on the next computing technology.

Now is the time for business leaders to start building small and nimble teams and exploring their creativity, which could potentially unlock new revenue streams. While there could be some downsides to being an early adopter of a new technology, the rewards for embracing spatial computing could far outweigh them.

Humans have been searching for the most natural way to interact with technology for decades. Spatial computing is just getting started, and business leaders have a chance to experiment and learn while building with an eye toward the development of the next computing platform.



TAKEAWAYS

The Apple Vision Pro marks the biggest experiment yet in the field of “spatial computing,” which proponents believe will be on par with the emergence of mobile computing. There are clear barriers to entry, as well as challenges and rewards for early adopters. But what are companies really doing with Apple’s new hardware, and what do they hope to gain from these investments? This article examines four cases.

- # Lowe’s believes that its experiments with spatial computing may help its customers manage the complexity of design and make better home improvement decisions.
- # e.l.f. Cosmetics sees spatial computing as part of the brand’s journey and vision to disrupt norms, shape culture, and connect communities.
- # The PGA Tour’s immersive “PGA Tour Vision” app has a goal of growing its fanbase and reaching fans in a new manner.
- # Hanifa’s Vision Pro app is still in development, but it expects the technology to become an essential piece of its ethos of pushing fashion’s boundaries.

NOTE

1. Sam Barnard, “Apple Predicted to Ship 350,000 Apple Vision Pro’s in First Year of Launch,” Techopedia, February 5, 2024, <https://www.techopedia.com/apple-predicted-ship-350000-apple-vision-pros-first-year-launch>.

Adapted from content posted on hbr.org, February 9, 2024 (product #H080Z2).

4

THREE DRIVERS OF CHINA'S BOOMING ELECTRIC VEHICLE MARKET

by Chengyi Lin

When it comes to the electric vehicle (EV) market, China is leading the charge ahead of traditional automotive juggernauts like Germany and Japan. China's new EV sales increased by 82% in 2022, accounting for nearly 60% of global EV purchases. This greatly surpasses that of the United States, Norway, and other Scandinavian nations that were early adopters of EVs.

According to a report by the International Energy Agency, more than half of the electric cars on roads worldwide are found in China. The country was also responsible for 35% of global EV exports in 2022. In fact, Chinese automaker BYD sold more electric vehicles than Tesla in Q4 2023.

How did China get to this point? What did Chinese car makers do differently, and what can companies looking to scale up their innovations learn from their approach? A sizable internal market and favorable government policies set the stage for this significant rise. However, other nations have implemented similar policies, yet haven't been as effective in accelerating mass market EV adoption.

This article outlines three key reasons for the growth of China's EV sector: experimenting in adjacent industries, encouraging operational solutions, and doubling down on core technology.

Experiment in Adjacent Industries

China began its EV drive later than the United States. While both countries had similar policies to incentivize companies and consumers, Chinese companies did not play the game directly.

Tesla CEO Elon Musk leveraged the media to position the brand as a pioneer in EV, which helped it break into the California market and quickly draw a strong national and global following. Instead of taking a similar "big banner" approach, directly targeting the auto industry, Chinese automakers BYD

and Geely remained under the radar and quietly experimented in their early stages. They kickstarted their EV development by focusing on adjacent industries—namely, electric buses and motorcycles. These products are less visible than cars, yet present unique challenges that are ripe for automakers to address. What they learned by tackling these challenges ultimately contributed to their EV manufacturing strategy.

For instance, buses are heavier and carry more passengers than commercial sedans. Additionally, most buses are operational for about 18 hours each day. They therefore have greater battery power and storage requirements. And more powerful batteries take longer to charge. By targeting an adjacent industry, BYD began pushing the boundaries of battery technology as early as 2009. BYD featured electric buses as its entry product into North American markets. It first sold electric buses as fleet vehicles in 2013, before supplying them to the Los Angeles Metro system in 2015. BYD electric buses are now also prevalent in South American markets.

Geely operates within another adjacent industry that presents different challenges: motorbikes, which require lighter and more portable batteries than cars. Experimenting in this area allowed the company to become a leading battery producer.

Taking an indirect path, these two Chinese companies have become EV giants by innovating on the two extremes of battery technology, which is core to EV production.

Encourage Operational Solutions

The second reason China's EV market has soared is that early innovators recognized the operational challenges that EVs presented and worked collaboratively with local groups to find solutions. Government policy may aim to accelerate the adoption of new technologies like EVs, but such innovations often introduce operational hurdles.

For example, many European countries such as the Netherlands were quick to encourage EV adoption by implementing registration tax incentives and rebates. However, research found that interest in purchasing full EVs remained very low among taxi drivers. A possible explanation for this lack of uptake could be operational challenges of full EVs, including the short driving range and long charging time. These concerns have tended to overshadow the environmental benefits and other strengths of EVs, such as having a quieter engine and not requiring regular oil or battery changes.

How did China overcome these operational hurdles? In 2009, China's government put in place a similar policy, to subsidize the purchase of hybrid and electric cars and buses in 10 cities. According to the policy, the per unit subsidies for passenger cars ranged from RMB4,000 (roughly \$500 in U.S. currency) to RMB60,000 (roughly \$8,000). But China went beyond subsidies.

In these 10 major cities, such as Beijing and Xi'an, Chinese EV producers worked closely with taxi companies to devise operational solutions that would improve core battery technologies. For example, EV companies didn't merely map out the locations for charging stations; more importantly, they tested various scheduling options for battery charging that matched the current performance level of fully electric and hybrid vehicles.

EVs equipped with the best battery technology can run for up to eight hours in the inner city. In China, taxi companies operating electric or hybrid vehicles typically have two fleets of cars—one for morning and one for evening shifts. The morning shift ends around 6–7 p.m., after the workday but before the evening rush. This enables the morning fleet to be charged after 8 p.m., avoiding the window of heavy industrial power consumption. The evening fleet returns for charging around 2–3 a.m., which is also within the period of lower power consumption for a city's grid.

This new schedule, designed jointly by Chinese EV producers and taxi companies, not only addresses the battery constraints of EVs but also helps to flatten the consumption curve of a city's power grid.

Double Down on Core Technology

European and U.S. automakers have historically had a strong foothold on the core technology for combustion engines. The Chinese auto industry fell significantly behind both regions, as well as Japan, in this area. But in 2002, Chinese automakers estimated that battery costs would comprise between 30% to 40% of the total manufacturing cost of a fully electric vehicle. This meant that there was a window of opportunity for newcomers to leapfrog the competition by focusing on the technology that powers this central component.

Coincidentally, the Chinese EV industry enjoys a proximity to many critical raw material supplies. For example, in 2022 China accounted for 70% of global production of rare earth, a central component for battery production. This means that Chinese battery companies control the bottleneck position of the supply chain, which can provide both positional advantages for them to develop new battery technologies and negotiation power with suppliers beyond batteries.

Indeed, Chinese companies collaborated broadly—with other automakers as well as technology companies—to strengthen their capabilities in terms of EV manufacturing.

When BYD began its EV journey, it shifted from manufacturing mobile phone batteries (it supplied both Nokia and Motorola) to manufacturing automotive batteries through Yadi Electronics, now part of BYD. Via acquisition of Qinchuan Machinery Works, a small car manufacturing company, BYD set up a new automotive division in 2002 and began making cars. BYD then collaborated with Daimler and Toyota to gain knowledge of EV manufacturing in exchange for sharing its own knowledge of battery manufacturing technologies. BYD also now works closely with Foshan Plastics Group on optoelectronics, which is the building of electronic sensors that detect and control lights. In 2018, BYD partnered with Chinese technology giant Baidu to scale up the software capability and service capacity of its EVs for the mass market. The mutually beneficial agreement equipped BYD EVs with Baidu Map and intelligent-driving software, while enabling Baidu to join BYD's open-source platform D++ and gain hardware knowledge and access to data.

Similarly, Geely set up an ecosystem that encompassed everything from low-orbit satellites to smart hardware to collect and monitor data that could potentially improve EV battery performance. It also partnered with Baidu, which builds the cloud-based software that controls its vehicles, on a joint venture (Jidu Auto) that aims to produce intelligent EVs. Geely then acquired Australian automatic transmission manufacturer Drivetrain Systems International, which supplies Ford, Maserati, and Chrysler, among others. It further acquired Volvo and Lotus, among other automakers, and has partnered with five more—including Daimler Smart—on other joint ventures.

Through these critical partnerships and acquisitions, Chinese automakers have charged up their development of peripheral components for EVs and accelerated their go-to-market speed. This organic and inorganic approach to ecosystem building allowed BYD and Geely to quickly and effectively orchestrate complementary assets around their core focus—battery technologies—which in turn helped them emerge as two leading EV manufacturers in China.

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Chinese companies have fueled the acceleration of their country's EV sector by creating innovative solutions to both technological and operational challenges around manufacturing and adoption. They have developed a deep understanding of what's required to move this industry forward.

At the same time, despite rising EV adoption, European automakers seem to have a hard time transitioning from internal combustion engine (ICE) cars to a more balanced portfolio of ICE, hybrid, and EV. A leading automotive executive attending one of my classes commented, "Our only job is to deliver our quarterly sales numbers [of traditional cars]. Someone else is dealing with innovation and mobility in the headquarters somewhere."

The next chapter for Chinese EV companies is international expansion. It will be interesting to see whether they succeed here. Having the core ingredients—strong battery technology, a firm hold over the battery supply chain, and operational advantages—is not enough to guarantee they remain a market leader in EV production.

Ultimate success lies beyond the product itself. Chinese EV companies need to repeat the same approach outlined above to learn about the global market—from channels to competition, from consumer behavior to infrastructure. For example, both the United States and Europe are tightening EV subsidies to benefit only local manufacturers. Can Chinese EV companies build local manufacturing or assembly plants too? What benefits would that bring? Many European markets are still building charging infrastructures. Can Chinese EV companies effectively participate in this process? Tesla, for example, had to collaborate with local body shops in France for EV repairs. Can Chinese EV companies build post-sales service networks in the U.S. and European markets? Experimenting with more operational solutions will surely help China drive EV adoption even further, into the global markets.



TAKEAWAYS

More than half of the EVs on roads worldwide are found in China. Companies looking to scale up their innovations—whether in the automotive industry or elsewhere—should understand three key decisions that accelerated the growth of China's EV sector.

- # **Experimenting in adjacent industries.** Chinese automakers BYD and Geely kickstarted their EV development in electric buses and motorcycles.
- # **Encouraging operational solutions.** In addition to generous government subsidies for EV purchasers, Chinese EV manufacturers worked with taxi companies to place charging stations at locations that matched the current performance level of battery technology.
- # **Doubling down on core technology.** Chinese companies collaborated broadly—with other automakers as well as other technology companies—to develop ecosystems and partnerships that strengthened their battery manufacturing capabilities.

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